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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

EE3252-1- POWER ELECTRONICS & DRIVES FOR ELECTRIC VEHICLES
(For EE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. In advanced HEVs the Air Conditioners are driven by
 - A) Diesel engines
 - B) Gasoline engines
 - C) Fuel cells
 - D) Electric motors
2. DC-DC converters are required in HEV
 - A) To charge auxiliary battery
 - B) To ensure the output voltage is constant
 - C) To power-up main motor
 - D) To charge main battery
3. The parasitic resistance of the capacitor will cause the _____ in a DC-DC converter
 - A) Discontinuous conduction mode
 - B) Ripple in source current
 - C) Ripple in capacitor current
 - D) Continuous conduction mode
4. Following is the constraint that needs to be considered in design of rectifiers used in HEVs compared to rectifiers used in general applications
 - A) Diode drops
 - B) Filters
 - C) Forward bias voltage of diodes
 - D) Impedance of the generator
5. The Expression for voltage ripple in diode rectifier is, -----
 - A) $\Delta V_0 = \frac{V_0 \pi}{2CR\omega}$
 - B) $\Delta V_0 = \frac{V_0}{2CR\omega}$
 - C) $\Delta V_0 = \frac{V_0 R}{2CR\pi\omega}$
 - D) $\Delta V_0 = \frac{V_0 \omega}{2CR\pi}$
6. For a typical passenger vehicle (car and SUV), the average energy consumption is approximately -----per mile.
 - A) 15–30Wh
 - B) 30–60 Wh
 - C) 150–300Wh
 - D) 300–600 Wh
7. The battery onboard in a PHEV needs to be charged
 - A) Main battery
 - B) Small generator fixed to motor drive
 - C) from grid using onboard or offboard charger
 - D) from grid using offboard charger only
8. A PFC stage is usually placed between
 - A) Supply and rectifier stage
 - B) the rectifier and the DC–DC stage
 - C) dc-dc stage and motor/ battery
 - D) Auxiliary to power source
9. When compared to PM motors, induction motors have
 - A) lower efficiency and less torque density
 - B) higher efficiency and high torque density
 - C) lower efficiency and high torque density
 - D) higher efficiency and less torque density
10. Maximum speed achievable by a induction motor connected to 50Hz supply
 - A) 1500rpm
 - B) 3000rpm
 - C) 750rpm
 - D) 6000rpm

11. Field-oriented control (FOC) enables an AC induction motor to be controlled like a _____ by using complex transformations
 A) Synchronous motor B) Switched Reluctance motor
 C) DC motor D) BLDC motor
12. When operated as a motor, the three-phase windings of PM motors are supplied with a trapezoidal form of current for _____ motor.
 A) Brushless DC B) Synchronous AC
 C) Induction D) Switched reluctance
13. FEA refers to
 A) Finite Element Analysis B) Finite Efficiency Analysis
 C) Fixed Efficiency Analysis D) Field Element Analysis
14. For a 6/4 pole-paired DSPM machine, if the width of the rotor pole equals that of the stator and their width is one-half of the pitch, then _____ torque will reach its minimal value
 A) Shaft B) cogging
 C) Breaking D) crawling
15. R_{ag} of the thermal model of the motor refers to
 A) convection thermal resistance of the air gap B) radial thermal resistance of the air gap
 C) convection thermal resistance of the armature circuit D) convection thermal resistance of the stator yoke
16. For low-frequency operations, the stator voltage must be increased to compensate _____
 A) the voltage drop across the stator resistance and inductance B) Copper loss in the stator and rotor windings
 C) Core loss D) saturates of rotor winding
17. The main purpose of using multiple power sources in HEVs is to
 A) Increase fuel cost B) Handle variable load demands efficiently
 C) Complicate system design D) Reduce driver comfort
18. What does the cost function represent in energy management?
 A) Vehicle maintenance cost B) Mathematical relation combining efficiency and penalties
 C) Battery price variation D) Cost of driver training
19. The state of charge (SOC) of a battery represents
 A) The temperature of the battery B) The voltage drop across the terminals
 C) The remaining capacity of the battery compared to its full charge D) The rate of discharge current
20. Dynamic programming in HEV energy management is based on
 A) Predicting the entire drive cycle B) Random selection of parameters
 C) Manual tuning by driver D) Disconnecting auxiliary loads

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the role of power electronics in series HEV with required block diagram. Discuss the issues to be addressed in the design of HEV power electronics circuits.	08	L*2	1	1,2
	b) Explain the PWM rectifier used in EVs for battery charging applications. How regenerative braking in the constant power region can be achieved? Analyze with suitable equivalent circuit and phasor diagram at high speed in the constant power region.	08	L4	2	1,2
2.	a) Explain the modes of operation of bidirectional DC-DC converter under heavy loaded conditions. Give the equation for secondary voltage in terms of the input voltage.	07	L2	1	1,2
	b) Explain the working of continuous and discontinuous mode of Full-Bridge DC-DC Converter.	06	L2	2	1,2
	c) Analyze the need for PFC when designing PHEV charger	03	L4	2	1,2

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|-------|--|----|----|---|-----|
| 3. a) | Explain the application of buck converter in HEV. Explain the buck converter and the design of filter elements used in the converter to ensure minimum ripple. | 06 | L2 | 1 | 1,2 |
| b) | Explain the challenges in maintaining Constant Torque Range and Power Capabilities when inverter-motor is directly connected to the battery. | 06 | L2 | 2 | 1,2 |
| c) | Analyze the Impact of thermal interface material conductivity on temperature difference. | 04 | L4 | 2 | 1,2 |

Unit – II

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|-------|--|----|----|---|-----|
| 4. a) | Explain the variable voltage variable frequency control of 3-phase induction motor. Discuss about the losses in induction motor due to PWM Supply | 06 | L2 | 3 | 1,2 |
| b) | What are the parameters to decide Sizing of Magnets in PM Motors? Explain | 05 | L2 | 3 | 1,2 |
| c) | Explain the concept of Thermal Resistance of the Air Gap and Rotor Core | 05 | L2 | 4 | 1,2 |
| 5. a) | Explain the construction and working principle of interior permanent magnet motor. Draw the phasor diagram and explain the concept of field weakening. | 08 | L2 | 3 | 1,2 |
| b) | Explain the Design and Sizing of Traction Motors. What are the factors that decide the speed ranges of motors used in traction drives. | 08 | L2 | 4 | 1,2 |
| 6. a) | Explain the construction and principle of operation of Doubly Salient Permanent Magnet Machines. Explain the existence of cogging in DSPM | 08 | L2 | 3 | 1,2 |
| b) | Draw the thermal Equivalent circuit model of traction drives and explain the components of the same. | 08 | L2 | 4 | 1,2 |

Unit – III

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|-------|---|----|----|---|-----|
| 7. a) | Explain the overall power management control implementation interface for a vehicular system | 08 | L2 | 5 | 1,2 |
| b) | Explain and differentiate the nature of electrical system characteristics for battery, fuel cell and generators. Discuss on the torque generated by EV when these sources are used. | 08 | L2 | 5 | 1,2 |
| 8. a) | Which are the two perspectives of power management and control in vehicular energy management systems? Explain | 08 | L2 | 5 | 1,2 |
| b) | Explain different methodologies for optimization in vehicular energy management. What is Cost Function Optimization? | 08 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
